

Lecture 13: Public-Key Encryption Schemes (Cont.)

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Warning: This document is a rough draft, so it may contain bugs. Please feel free to email me with corrections.

Recap:

In the last lecture, we saw how to construct a CPA-secure public-key encryption scheme from any trapdoor injective one-way function family.

Today:

We will see constructions of trapdoor one-way functions from number theoretic assumptions.

First, let us recall the definition of trapdoor one-way functions and how they can be used to construct a public-key encryption scheme.

Definition 1. A trapdoor injective one-way function family consists of a PPT key generation algorithm Gen , that takes as input 1^λ and outputs a (public) hash key hk together with an associated (secret) trapdoor key td , and a family of functions $F_{\text{hk}} : \mathcal{D}_{\text{hk}} \rightarrow \mathcal{R}_{\text{hk}}$. The following properties are required to hold:

- **Efficiently computable:** There is a poly-time algorithm $\mathcal{A}$ such that for every $(\text{hk}, \text{td}) \in \text{Gen}(1^\lambda)$ and for every $x \in \mathcal{D}_{\text{hk}}$, $\mathcal{A}(\text{hk}, x) = F_{\text{hk}}(x)$.
- **Efficiently sampleable:** There is a PPT algorithm $\mathcal{D}$ such that for any $(\text{hk}, \text{td}) \leftarrow \text{Gen}(1^\lambda)$,

$$\mathcal{D}(\text{hk}) \equiv U_{\mathcal{D}_{\text{hk}}}$$

Namely $\mathcal{D}(\text{hk})$ outputs an element in $\mathcal{D}_{\text{hk}}$ whose distribution is statistically close to the uniform distribution over the domain $\mathcal{D}_{\text{hk}}$.

- **Trapdoor invertible:** There is a poly-time inversion algorithm $\mathcal{I}$ such that for any $(\text{hk}, \text{td}) \in \text{Gen}(1^\lambda)$ and any $x \in \mathcal{D}_{\text{hk}}$

$$\mathcal{I}(\text{td}, F_{\text{hk}}(x)) = x$$

- **One-way:** For every polys-size $\mathcal{A}$ there exists a negligible function μ such that for every $\lambda \in \mathbb{N}$,

$$\Pr[\mathcal{A}(\text{hk}, F_{\text{hk}}(x)) = x] \leq \mu(\lambda)$$

where the probability is over $(\text{hk}, \text{td}) \leftarrow \text{Gen}(1^\lambda)$ and over $x \leftarrow \mathcal{D}_{\text{hk}}$.

The corresponding public-key encryption scheme is defined as follows, where $\text{HCP}(x; r)$ is any hardcore predicate such as the Goldreich Levin (randomized) hardcore predicate

$$\text{HCP}(x; r) = x \cdot r.$$

- $\text{Gen}(1^\lambda)$ outputs $(\text{pk}, \text{sk}) = (\text{hk}, \text{td})$.
- $\text{Enc}(\text{hk}, m) = (x, F_{\text{hk}}(x), r, \text{HCP}(x; r) \oplus m)$.
- $\text{Dec}(\text{td}, y, r, c)$ uses td to obtain x which is the inverse of y , and then use (x, r) to unmask c and obtain $m = \text{HCP}(x; r) \oplus c$.

Security relies on the fact that by definition of a hardcore predicate,

$$(\text{hk}, F_{\text{hk}}(x), r, \text{HCP}(x; r)) \approx (\text{hk}, F_{\text{hk}}(x), r, U)$$

where $U \leftarrow \{0, 1\}$.

Number-Theory Review

The constructions we will see are based on number theory, but this time over the group

$$\mathbb{Z}_N^* = \{x : \text{GCD}(x, N) = 1\}.$$

where N is a product of two large primes $p, q \leftarrow \{0, 1\}^\lambda$.

The Chinese Remainder Theorem (CRT) The CRT theorem tells us that for $N = p \cdot q$, it holds that $c = a \cdot b \bmod N$ if and only if

$$c = a \cdot b \bmod p \quad \wedge \quad c = a \cdot b \bmod q.$$

Furthermore, for any $a \in \mathbb{Z}_p^*$ and $b \in \mathbb{Z}_q^*$, there is a unique *efficiently computable* $c \in \mathbb{Z}_N^*$ such that

$$c = a \bmod p \quad \wedge \quad c = b \bmod q.$$

In other words, the CRT theorem asserts that $\mathbb{Z}_N^*$ is isomorphic to $\mathbb{Z}_p^* \times \mathbb{Z}_q^*$, with the isomorphism given by

$$x \rightarrow (x \bmod p, x \bmod q).$$

Claim 1. Given any primes p, q and any $a, b \in \mathbb{N}$ that are co-prime to p and q , respectively, one can efficiently compute $c \in \mathbb{Z}_N^*$, for $N = p \cdot q$, such that

$$c \bmod p = a \quad \wedge \quad c \bmod q = b$$

Proof. Given any primes p, q and any $a, b \in \mathbb{N}$, co-prime to p and q , respectively, compute $c \in \mathbb{Z}_N^*$, as follows.

- Compute p^{-1} which is the inverse of p modulo q .
- Compute q^{-1} which is the inverse of q modulo p .
- Output $c = q \cdot q^{-1} \cdot a + p \cdot p^{-1} \cdot b \bmod N$.

Remark. The notation above is a bit misleading since $p \cdot p^{-1} \neq 1 \bmod N$, and similarly for q . Moreover, neither p nor q have an inverse modulo N . Crucially, the inverses p^{-1} and q^{-1} were defined over q and p , respectively.

It is easy to see that

$$c \bmod p = a \quad \wedge \quad c \bmod q = b,$$

as desired. □

Notation We denote by (a, b) the CRT representation of an element $c \in \mathbb{Z}_N^*$. Namely (a, b) refers to the unique $c \in \mathbb{Z}_N^*$ such that

$$c \bmod p = a \quad \wedge \quad c \bmod q = b,$$

Warmup Construction: Rabin's Function

As a warmup we present Rabin's function family [2]. It is not quite what we want, since it is not injective. We will later elaborate on how one can convert this trapdoor family into an injective one.

The Construction

- $\text{Gen}(1^\lambda)$:
 1. Choose two random primes $p, q \leftarrow \{0, 1\}^\lambda$.
 2. Compute $N = p \cdot q$.
 3. Output $\text{hk} = N$ and $\text{td} = (p, q)$.
- $F_N(x) : \mathbb{Z}_N^* \rightarrow \mathbb{Z}_N^*$, where

$$F_N(x) = x^2 \bmod N.$$

Definition 2. Factoring is said to be hard if for every poly-size $\mathcal{A}$ there exists a negligible μ such that

$$\Pr[\mathcal{A}(N) = (p, q)] \leq \mu(\lambda)$$

where the probability is over randomly chosen primes $p, q \leftarrow \{0, 1\}^\lambda$ and where $N = p \cdot q$.

Claim 2. Rabin's function family is a trapdoor (non-injective) one-way function family (where the inversion algorithm finds all preimages) assuming the hardness of Factoring.

Proof. We prove that this function family satisfies all the conditions from Definition 1 except the injectiveness.

- **Efficiently computable:** It is easy to see that given $hk = N$ and given any $x \in \mathbb{Z}_N^*$ it is easy to compute $F_N(x) = x^2 \bmod N$.
- **Efficiently sampleable:** One can efficiently sample an element from $\mathbb{Z}_N^*$ uniformly at random, by repeatedly sampling a random element in $\{1, \dots, N-1\}$ until we sample x such that $\gcd(x, N) = 1$.^{1,2}
- **Trapdoor invertible:** Given $td = (p, q)$ and given $y = F_N(x) = x^2 \bmod N$, we will use the Chinese Remainder Theorem (CRT), compute all four square-roots of y modulo N , as follows:

1. Compute $a \in \mathbb{Z}_p^*$ such that $a^2 = y \bmod p$.

This is done by computing $\alpha = 2^{-1} \bmod (p-1)$, and setting $a = y^\alpha \bmod p$.

Note that a and $p-a$ are the only two square roots of y modulo p . This follows from the fact that p is prime and hence $\text{GF}[p]$ is a field.

2. Compute b such that $b^2 = y \bmod q$.

This is done in a similar manner, and as above, β and $q-\beta$ are the only two square roots of y modulo q .

3. For every one of the four elements

$$(a, b), (a, q-b), (p-a, b), (p-a, q-b) \in \mathbb{Z}_p^* \times \mathbb{Z}_q^*$$

output the corresponding elements $c_1, c_2, c_3, c_4 \in \mathbb{Z}_N^*$.

4. **One-way:** We will show that if the one-way condition does not hold then we can break the hardness of Factoring assumption. Specifically, suppose that there exists a poly-size $\mathcal{A}$ and a non-negligible ϵ such that

$$\Pr[\mathcal{A}(N, y) = x : x^2 = y \bmod N] \geq \epsilon(\lambda)$$

¹ Note that with overwhelming probability, the first sample will satisfy this condition.

² Note that $\gcd$ is an efficiently computable function, via Euclid's the extended $\gcd$ algorithm.

where the probability is over randomly chosen primes $p, q \leftarrow \{0, 1\}^\lambda$, and setting $N = p \cdot q$, and over a random element in the image of F_N .

We will construct a poly-size algorithm that breaks the hardness of Factoring. Given $N = p \cdot q$, the algorithm $\mathcal{B}$ does the following:

- (a) Choose a random element $x \leftarrow \mathbb{Z}_N^*$.
- (b) Compute $y = x^2 \bmod N$.
- (c) Compute $x' = \mathcal{A}(N, y)$.
- (d) Compute $d = \gcd(x, x')$.
- (e) If $d = 1$ then abort, and otherwise output $(d, N/d)$.

We next argue that $\mathcal{B}$ successfully factors N with probability $\frac{\epsilon(\lambda)}{2}$. To this end denote by Succ the event that $\mathcal{A}$ is successful at finding a preimage.

$$\begin{aligned} \Pr[\mathcal{B}(N) = (p, q)] &= \\ \Pr[\mathcal{B}(N) = (p, q) \mid \text{Succ}] \cdot \Pr[\text{Succ}] &+ \Pr[\mathcal{B}(N) = (p, q) \mid \neg \text{Succ}] \cdot \Pr[\neg \text{Succ}] \geq \\ \Pr[\mathcal{B}(N) = (p, q) \mid \text{Succ}] \cdot \Pr[\text{Succ}] &\geq \\ \Pr[\mathcal{B}(N) = (p, q) \mid \text{Succ}] \cdot \epsilon(\lambda) &= \\ \frac{1}{2} \cdot \epsilon(\lambda) \end{aligned}$$

where the last equation follows from the fact that y has four preimages modulo N :

$$(a, b), (a, q - b), (p - a, b), (p - a, q - b),$$

and $\mathcal{A}$ has no information about the preimage x chosen by $\mathcal{B}$. Hence, the output would be identical if first $\mathcal{A}$ would output a preimage x' and then $\mathcal{B}$ would choose a random preimage x . Thus, with probability $1/2$ the two preimages will be equal modulo one of the primes and not equal modulo the other, in which case indeed $\gcd(x, x')$ will output one of the primes.

□

Rabin's construction is very nice but it is not injective, and hence it is not clear how it can be used to construct a public-key encryption scheme. Note that we can easily make it injective by restricting its domain to be: Let

$$\mathbb{QR}_N = \{x^2 : x \in \mathbb{Z}_N^*\}.$$

We can define

$$F_N : \mathbb{QR}_N \rightarrow \mathbb{QR}_N$$

in which case it is a permutation! The problem is that we cannot efficiently sample from $\mathbb{QR}_N$ given only N . The reason is that in order to check if $x \in \mathbb{QR}_N$ one needs the trapdoor p, q . Note that $x \in \mathbb{QR}_N$ if and only if $x \in \mathbb{QR}_p$ and $x \in \mathbb{QR}_q$.

Nevertheless this construction turned out to be extremely useful, and has led to the Goldwasser-Micali encryption scheme [1], which was the first public-key encryption scheme that was proven to be secure!

The Goldwasser-Micali Public-Key Encryption Scheme

- $\text{Gen}(1^\lambda)$:
 1. Sample random primes $p, q \leftarrow \{0, 1\}^\lambda$, such that

$$(p \bmod 4 = 3) \wedge (q \bmod 4 = 3).$$

This choice of p, q ensures that (-1) is not a quadratic residue modulo p or modulo q .

This is the case since let $x \in \mathbb{Z}_p^*$ be any generator then $-1 = x^{\frac{p-1}{2}}$. Note that this would be a quadratic residue only if $\frac{p-1}{2}$ was even modulo $p-1$, i.e., if there exists $a \in \mathbb{N}$ such that

$$2a = \frac{p-1}{2} \bmod (p-1)$$

which holds only if $p \equiv 1 \pmod{4}$.

2. Let $N = p \cdot q$.
 3. Output $\text{pk} = N$ and $\text{sk} = (p, q)$.
- $\text{Enc}(\text{pk}, m)$:
 1. Sample a random $x \leftarrow \mathbb{Z}_N^*$.
 2. Compute $y = x^2 \bmod N$.
 3. Output $\text{ct} = (-1)^m \cdot y$.
 - $\text{Dec}(\text{sk}, \text{ct})$:
 1. Output $m = 0$ if $y \in \mathbb{QR}_N$ and output $m = 1$ otherwise.

Security follows from the fact that it is hard to distinguish random elements in $\mathbb{QR}_N$ from random elements that are not quadratic residues modulo p or q .

Remark. It turns out that it is easy to distinguish random elements in $\mathbb{QR}_N$ from random elements in $\mathbb{Z}_N^* \setminus \mathbb{QR}_N$. But we believe that it is hard to distinguish between random elements in $\mathbb{QR}_N$ from random elements in $\{x \in \mathbb{Z}_N^* : (x \bmod p) \notin \mathbb{QR}_p \wedge (x \bmod q) \notin \mathbb{QR}_q\}$. This

is called the Quadratic Residue assumption and is precisely what is needed in order to prove the security of the Goldwasser-Micali encryption scheme.

RSA Construction of a Trapdoor Permutation

We now present the RSA function, due to Rivest, Shamir, and Adleman [3]. The RSA trapdoor function came before Rabin's function, but it is convenient to think of it as a modification of Rabin's function, that converts it from a trapdoor function into a trapdoor permutation.

Recall that Rabin's function $F_N(x) = x^2 \bmod N$ had the problem that each image has 4 preimages. This results from the fact that in $\mathbb{Z}_p^*$ and $\mathbb{Z}_q^*$ every square has two square roots, and thus by the CRT theorem, in $\mathbb{Z}_N^*$ each square has four square roots.

There is a natural fix to this! Instead of taking $F_N(x) = x^2 \bmod N$, define it to be

$$F_{N,e}(x) = x^e \bmod N$$

where e is any element that is co-prime to $p-1$ and $q-1$. The point is that if e is co-prime to $p-1$ and $q-1$ then $F_{N,e} : \mathbb{Z}_N^* \rightarrow \mathbb{Z}_N^*$ is a permutation. The reason is that if e is co-prime to $p-1$ then there exists d such that $d \cdot e = 1 \bmod (p-1)$ and hence if

$$x_1^e = x_2^e \bmod N$$

then

$$x_1^e = x_2^e \bmod p \quad \wedge \quad x_1^e = x_2^e \bmod q.$$

Note that

$$(x_1^e)^d \bmod p = x_1^{1+e \cdot d \cdot (p-1)} \bmod p = x_1,$$

and the same holds for q and for x_2 . This implies that

$$x_1 = x_2 \bmod p \quad \wedge \quad x_1 = x_2 \bmod q$$

which by the CRT theorem implies that $x_1 = x_2$.

One way of choosing such e is setting $e = 3$ and choosing random p, q such that $p \bmod 3 = 2$ and $q \bmod 3 = 2$. This ensures that 3 is co-prime to $p-1$ and $q-1$.

This is exactly the RSA trapdoor permutation! The next question to ask is whether this is still one-way assuming the hardness of Factoring? The sad answer is that we don't know. Instead we make a new assumption, referred to as the RSA assumption.

The RSA Assumption: For every poly-size $\mathcal{A}$ there exists a negligible function μ such that for every $\lambda \in \mathbb{N}$,

$$\Pr[\mathcal{A}((N, e), F_{N, e}(x)) = x] \leq \mu(\lambda)$$

where the probability is over randomly chosen $p, q \leftarrow \{0, 1\}^\lambda$ such that $p \bmod 3 = 2$ and $q \bmod 3 = 2$, and over $x \leftarrow \mathbb{Z}_N^*$ where $N = p \cdot q$ and $e = 3$.

Under this assumption the one-wayness holds by definition!

Remark. The original proposed RSA encryption scheme worked with message space $\mathcal{M}_\lambda = \mathbb{Z}_N^*$, and the encryption algorithm was defined to be

$$\text{Enc}((N, e), m) := m^e \bmod N.$$

However, this is clearly insecure, since it is deterministic! We can make it secure by using the trapdoor function with a hardcore predicate, as we saw in the last lecture. Namely, let $\mathcal{M} = \{0, 1\}$ and

$$\text{Enc}((N, e), m) = (F_{N, e}(x), r, \text{HCP}(x; r) \oplus m).$$

We can take HCP to be the Goldreich-Levin hardcore predicate. But, it turns out that the least-significant bit $P(x) := x \bmod 2$ is a hardcore predicate for RSA. Namely, if there exists a poly-size adversary that guesses the least-significant bit of x with non-negligible advantage given $x^e \bmod N$, then there is a poly-size adversary that breaks the RSA Assumption with non-negligible probability.

So, one can use the following encryption algorithm:

$$\text{Enc}((N, e), m) = (F_{N, e}(x), (x \bmod 2) \oplus m).$$

References

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